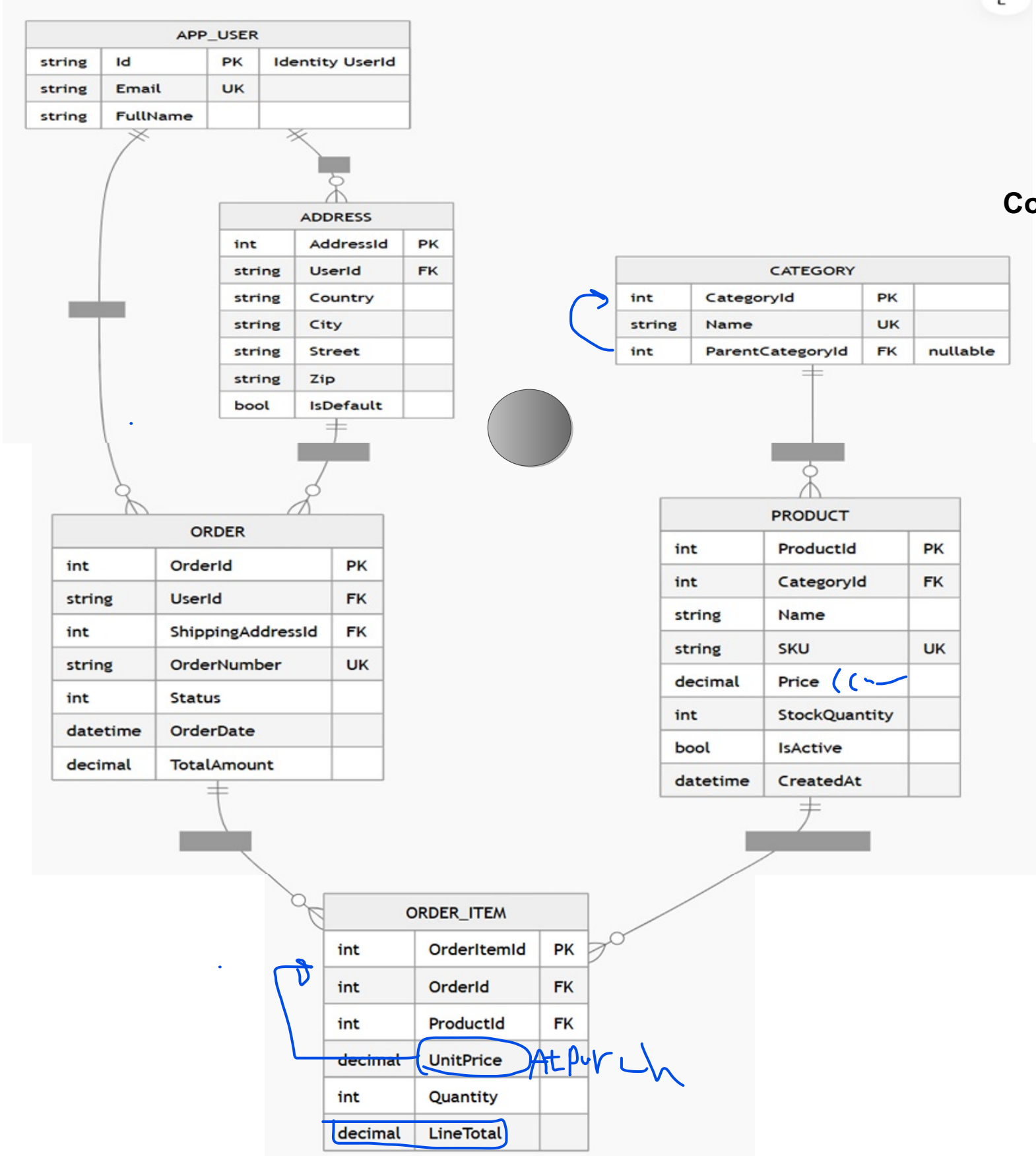

Conceptual relationships

- Category 1—∞ Product
- Customer (Identity User) 1—∞ Order
- Order 1—∞ OrderItem
- Product 1—∞ OrderItem
- Customer 1—∞ Address
- Order ∞—1 Address (shipping)

MVC Project Type and Structure

Project template

- **ASP.NET Core Web App (Model-View-Controller)**
- **Authentication: Individual Accounts** (ASP.NET Core Identity)
- **Database: SQL Server**
- **EF Core** with migrations - Code First

MVC Pages (Views) You Should Build

Public

- **Catalog/Index**: browse products + filter by category + search + pagination
- **Catalog/Details/{id}**
- **Cart/Index**
- **Cart/Add/{productId}**, **Cart/Update**, **Cart/Remove**
- **Checkout/Index** (address + confirm)
- **Orders/Index** (my orders)
- **Orders/Details/{id}**

Admin

- **Admin/Categories** CRUD
- **Admin/Products** CRUD
- **Admin/Orders** list + update status

Controllers and Responsibilities

CatalogController

- `Index(categoryId, q, sort, page)`
- `Details(id)`

CartController

- `Index()`
- `Add(productId)`
- `Update(itemId, qty)`
- `Remove(itemId)`
- Cart storage choice:
  - **Session-based cart** (simple)
  - Or DB cart tables (more advanced)

OrdersController

- `Checkout()` GET
- `Checkout(CheckoutVM)` POST
- `Index()` (customer orders)
- `Details(id)`

Admin Area Controllers

- Admin authorization: `[Authorize(Roles = "Admin")]`
- Products CRUD + category selection
- Orders status updates

ViewModels (Important in MVC)

Use ViewModels to avoid passing EF entities directly to views.

Examples:

- `ProductListVM` (items + paging + filter state)
- `ProductDetailsVM`
- `CartVM` (items + totals)
- `CheckoutVM` (selected address + cart summary)
- `OrderDetailsVM` (order header + items)
- .....

Checkout transaction (must be atomic)

During checkout POST:

1. Validate stock for all items
2. Create `Order`
3. Create `OrderItems`
4. Decrease product stock
5. Save changes inside one transaction

Notes

- Start with CRUD operations for all entities.
- Use view models.
- Use repositories.
- We will add authentication and authorization after authentication lectures.